Finite-Size Graph-RH for the GeoVac Hopf Graphs: Ihara Zeta, Bound-Crossing, and a Scope Boundary on Selberg-on-Hydrogen

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Abstract

We compute the Ihara zeta function of the two GeoVac Hopf graphs – the Fock-projected S^3 graph for the Coulomb problem [2] and the Bargmann–Segal S^5 graph for the 3D isotropic harmonic oscillator [5] – via the Ihara–Bass determinantal formula [10, 11] and the Hashimoto non-backtracking edge operator [12]. Four results emerge. (1) *Graph-Ramanujan at small sizes, bound-crossing at modest sizes*: every graph tested at small sizes (S^3 at $n_{\max} = 2, 3$; S^5 at $N_{\max} = 2, 3$) satisfies the Kotani–Sunada [13] graph-RH (Ramanujan) bound *strictly*, with deviations of the largest non-trivial Hashimoto eigenvalue from $\sqrt{q_{\max}}$ ranging from -0.198 (S^3, $n_{\max} = 3$) to -0.357 (S^5, $N_{\max} = 3$). A follow-on post-draft extension to $n_{\max}, N_{\max} \in \{4, 5\}$ establishes that three of four graph families *cross* the Kotani–Sunada bound at modest sizes ($V \approx 30$–60), with the signed deviation growing linearly in V; the graph-RH for GeoVac is therefore a *finite-size* statement, not an asymptotic one. (2) *Algebraicity of zeros*: every $\zeta_G(s)^{-1}$ factors as a product of integer-coefficient polynomials; no transcendental (π, $\zeta(2)$, $\zeta(3)$) enters the zero set. This sharpens Paper 24's π-free certificate [5] from the adjacency data to the Ihara zeta. (3) *Structural factorization, validated*: the S^3 Coulomb zeta decomposes per ℓ-shell because the Fock adjacency preserves ℓ; the S^5 $N_{\max} = 3$ zeta factors as $(s\pm1)^{23} \cdot P_{12}(s^2) \cdot P_{22}(s^2)$, and the $12 + 22$ dichotomy is *exactly* the $\mathbb{Z}_2$ $m_\ell \to -m_\ell$ block decomposition of the node-level Ihara–Bass matrix — the only sub-action of the Paper 25 [6] Hopf $U(1)$ that commutes with a real integer adjacency. The analogous $22 + 24$ decomposition holds for the Dirac-S^3 Rule B $n_{\max} = 3$ graph under $m_j \to -m_j$; this magnetic reflection is a fixed-point-free (Kramers) graph automorphism, whereas its chirality companion $\chi \to -\chi$ is not a graph symmetry at all — the two spinor-bundle chiralities differing in size by the discrete Dirac index $n_{\max}(n_{\max} + 1)$, which explains the failure of relativistic $\mathbb{Z}_2$ tapering. (4) *Scope boundary*: Fock's Wick rotation $S^3 \to H^3$ is mathematically clean [9] but produces no natural discrete quotient H^3/Γ from framework invariants; the Selberg-zeta direction on hyperbolic quotients of hydrogen is therefore structurally closed. The Ihara-to-classical-ζ bridge is additionally blocked by bipartiteness (odd-argument Dirichlet characters such as χ_{-4} have support disjoint from the Ihara walk-length spectrum); the spectral-side Dirichlet series of Paper 28 remains the open direction.

1 Introduction

The GeoVac framework [1, 2] computes quantum-mechanical observables from discrete graph Laplacians on compact manifolds. In the accumulated results of the project, six transcendental constants have appeared at positive integer arguments of the Riemann zeta and Dirichlet L family: π (calibration [4]); $\zeta(2) = \pi^2/6$ as a Dirichlet series of the Fock degeneracy lattice at the packing exponent $d_{\max} = 4$; $\zeta(3)$ in the Dirac sector at $s = d_{\max}$ and in flat-space QED sums [7]; Catalan's constant $G = \beta(2)$ and the Dirichlet L-value $\beta(4)$ from two-loop vertex parity on S^3 [7]; and the irreducible

$S_{\min} \approx 2.47994$ [7]. Every member of this tally lives at a positive integer argument of ζ: they are *periods*, not *zeros*. The Riemann Hypothesis is about the location of nontrivial zeros of ζ on the critical line $\Re s = \frac{1}{2}$. Periods and zeros are structurally different objects; the standard bridges (explicit formula, Euler product) move data across the divide but do not collapse the distinction.

This paper reports what happens when one approaches the RH-bridge question from inside GeoVac rather than from outside. We do not attempt to prove or disprove RH. We ask: does the framework contain, on its own, a graph-zeta or spectral-zeta object with an RH-type structure — an Euler product, a functional equation, and a critical locus that our zeros respect? The answer, for the two Hopf graphs the framework natively builds, is yes. Every tested GeoVac graph satisfies the Kotani–Sunada [13] Ramanujan bound (Section 4). The associated functional equation takes the irregular-graph reflection form $s \mapsto -s$ rather than the regular-graph form $s \mapsto 1/(qs)$ (Section 6). And the Ihara zeros are algebraic, consistent with Paper 24's π-free structure (Section 5).

We also address the natural companion question: could the Wick-rotated side of the Fock projection — scattering states on hyperbolic H^3 [9] — carry a Selberg zeta with proven RH analog? The answer is no: GeoVac invariants supply no selection principle for a discrete subgroup $\Gamma \subset \mathrm{SO}(3,1)$ (Section 7). This closes a door cleanly. The graph-zeta direction is structurally intrinsic; the Selberg direction requires external arithmetic input that GeoVac does not provide.

2 Ihara Zeta and Graph-RH

For a finite connected graph $G = (V, E)$ with $|V|$ vertices and $|E|$ edges, the *Ihara zeta function* [10] is the formal Euler product

$$\zeta_G(s) \;=\; \prod_{[C]} \left(1 - s^{\ell(C)}\right)^{-1}, \tag{1}$$

where $[C]$ ranges over equivalence classes of *primitive* closed walks in G (walks with no backtracking and no tail, considered up to cyclic rotation) and $\ell(C)$ is the walk length. The product converges for $|s| < 1/\rho(T)$ where T is the Hashimoto non-backtracking edge operator (Section 2.2).

2.1 Ihara–Bass determinantal formula

A direct enumeration of primitive closed walks is infeasible for graphs with more than a few dozen edges. The Ihara–Bass theorem [11] replaces the Euler product by a finite-dimensional determinant:

$$\zeta_G(s)^{-1} \;=\; \left(1 - s^2\right)^{r-c} \det\!\left(I - s\,A + s^2\,Q\right), \tag{2}$$

where A is the $|V| \times |V|$ adjacency matrix, Q is the diagonal matrix with entries $d_v - 1$ (where d_v is the degree of vertex v), c is the number of connected components, and $r = |E| - |V| + c = \beta_1$ is the first Betti number. The $(r - c)$ prefactor is the product-over-components generalization of Terras's connected formula [14, Thm. 2.3]; it reduces to the standard $(1 - s^2)^{r-1}$ when $c = 1$.

2.2 Hashimoto non-backtracking operator

The *Hashimoto edge operator* [12] T acts on the $2|E|$-dimensional space of directed edges of G: for each directed edge $(u \to v)$,

$$T[(u \to v)] \;=\; \sum_{w \neq u,\, w \sim v} (v \to w). \tag{3}$$

Equivalently T is $|T| = 2|E| \times 2|E|$, with entries $T_{e,f} = 1$ iff edge f begins where edge e ends and $f \neq e^{-1}$. Then

$$\zeta_G(s)^{-1} \;=\; \det\bigl(I - s\,T\bigr), \tag{4}$$

so the reciprocal Ihara zeros of G are exactly the nonzero eigenvalues of T.

2.3 Ramanujan graphs and graph-RH

A $(q+1)$-regular graph is *Ramanujan* if every non-trivial eigenvalue λ of the adjacency matrix satisfies $|\lambda| \leq 2\sqrt{q}$ [15, 16]. For irregular graphs the Ramanujan condition is stated on the Hashimoto spectrum: G is *(weakly) Ramanujan* if every non-trivial eigenvalue μ of T satisfies $|\mu| \leq \sqrt{q_{\max}}$, where $q_{\max} = \max_v(d_v - 1)$ [13, 14]. The non-trivial eigenvalues are those other than the Perron eigenvalue $\rho(T)$ and its complex-conjugate reflection.

The *graph Riemann Hypothesis* is the statement that ζ_G is Ramanujan. Equivalently, the Ihara zeros (reciprocals of T-eigenvalues) are confined to the critical annulus $1/\sqrt{q_{\max}} \leq |s| \leq 1$, outside of which only trivial zeros (from the Perron eigenvalue) appear. This is a *proven* theorem on specific graph families (the LPS-Margulis-Morgenstern constructions [15, 16]) and has an extensive literature; see Terras [14] for a textbook account.

3 The GeoVac Hopf Graphs

GeoVac builds discrete graphs on two natural geometries: the three-sphere S^3 (the Fock [8] projection for hydrogenic bound states, Paper 7 [2]) and the five-sphere S^5 (the Bargmann–Segal projection for the 3D isotropic harmonic oscillator, Paper 24 [5]). Both carry Hopf-fibration structure ($S^1 \hookrightarrow S^3 \to S^2$ and $S^1 \hookrightarrow S^5 \to \mathbb{CP}^2$) and are the canonical discretizations referenced by Paper 25 [6]. They are the GeoVac instance of the Marcolli–van Suijlekom gauge-network graph-spectral-triple framework [25] (with the Perez-Sanchez correction [26] on the continuum limit). We summarize their combinatorial data.

3.1 S^3 Coulomb graph

Nodes are hydrogenic quantum-number triples (n, ℓ, m) with $1 \leq n \leq n_{\max}$, $0 \leq \ell < n$, $|m| \leq \ell$. Edges are generated by the $\mathrm{SU}(2) \times \mathrm{SU}(1,1)$ ladder operators $L_\pm, T_\pm$, which shift $m \mapsto m \pm 1$ (at fixed n, ℓ) and $n \mapsto n \pm 1$ (at fixed ℓ, m), respectively. The adjacency rule therefore *preserves* ℓ, which forces the graph to split into disconnected components indexed by ℓ (see Section 5).

3.2 S^5 Bargmann–Segal graph

Nodes are SU(3) weights (N, ℓ, m_ℓ) with $N \geq 0$, $\ell \in \{N, N-2, N-4, \ldots, 0 \text{ or } 1\}$, $|m_\ell| \leq \ell$, truncated at $N \leq N_{\max}$. Edges are the S^5 dipole transitions with $\Delta N = \pm 1, \Delta \ell = \pm 1, \Delta m_\ell \in \{0, \pm 1\}$, carrying exact-rational squared matrix elements (Paper 24 [5] §III). The adjacency is *not* ℓ-preserving and the graph is connected at every tested $N_{\max}$.

3.3 Combinatorial summary

4 Main Result: the Ramanujan Property

Applying (2) and (4) to each graph gives the verdicts in Table 2.

Graph	V	E	c	β_1	$q_{\max}$
K_4 (sanity)	4	6	1	3	2
$K_{3,3}$ (sanity)	6	9	1	4	2
S^3 Coulomb, $n_{\max} = 2$	5	3	2	0	1
S^3 Coulomb, $n_{\max} = 3$	14	13	3	2	2
S^5 Bargmann–Segal, $N_{\max} = 2$	10	15	1	6	4
S^5 Bargmann–Segal, $N_{\max} = 3$	20	42	1	23	8

Table 1: Combinatorial data for the graphs tested. V: vertices. E: edges. c: connected components. $\beta_1 = E - V + c$: first Betti number. $q_{\max} = \max_v(d_v - 1)$: Ramanujan bound parameter. The $\beta_1 = 23$ at S^5 $N_{\max} = 3$ agrees with the Paper 25 Hodge computation of $\beta_1 = 110$ for the same graph family at $N_{\max} = 5$ (the growth is roughly quadratic in $N_{\max}$).

Graph	$\rho(T)$	$\max|\mu_{\text{nontriv}}|$	$\sqrt{q_{\max}}$	Deviation	Ramanujan?
K_4	2.0000	1.4142	1.4142	0.0000	YES (exact)
$K_{3,3}$	2.0000	1.4142	1.4142	0.0000	YES (exact)
S^3 C., $n_{\max} = 2$	0	0	1.0000	−1.0000	YES (trivial; forest)
S^3 C., $n_{\max} = 3$	1.3532	1.2157	1.4142	−0.1985	**YES**
S^5 BS, $N_{\max} = 2$	2.3572	1.7922	2.0000	−0.2078	**YES**
S^5 BS, $N_{\max} = 3$	3.9053	2.4716	2.8284	−0.3568	**YES**

Table 2: Ramanujan verdicts. $\rho(T)$ is the Perron eigenvalue of the Hashimoto operator (a trivial eigenvalue, not counted against the Ramanujan bound). $\max|\mu_{\text{nontriv}}|$ is the largest non-trivial Hashimoto eigenvalue. Ramanujan requires $\max|\mu_{\text{nontriv}}| \leq \sqrt{q_{\max}}$; the deviation $\max|\mu_{\text{nontriv}}| - \sqrt{q_{\max}}$ is strictly negative for every non-trivial tested graph.

Observation 1. *At the small (closed-form) cutoffs of this section ($n_{\max} \leq 4$ on S^3, $N_{\max} \leq 3$ on S^5), the largest non-trivial Hashimoto eigenvalue is strictly smaller than the Kotani–Sunada bound: these graphs are strictly sub-Ramanujan, their spectral gap exceeding the optimal Ramanujan gap of an equivalent-$q_{\max}$ regular graph. This is a* finite-size *statement: the larger-cutoff sweep of §8 (RH-D) shows three of four families (including S^3 Coulomb at $n_{\max} = 5$)* cross *the bound, so the property does not persist asymptotically.*

The strictness of the deviation is itself structurally interesting. The Alon–Boppana bound [18] asserts that for a sequence of graphs of bounded degree, the largest non-trivial eigenvalue asymptotes to the Ramanujan bound from below; graphs that achieve the bound asymptotically are in this sense *optimal*. The closed-form data of this section do not by themselves establish the asymptotic behavior of the deviation; the larger-cutoff sweep in Section 8 (RH-D) resolves it — the signed deviation crosses the bound for three of four families, so the graphs are *not* asymptotically Ramanujan (the graph-RH here is a finite-size statement).

5 Closed-Form Ihara Zetas and Structural Factorizations

Working in exact rational arithmetic via `sympy`, we obtain explicit closed forms for each graph's Ihara zeta.

5.1 S^3 Coulomb at $n_{\max} = 3$: per-ℓ-shell decomposition

$$\zeta_G(s)^{-1} \;=\; -(s-1)^2(s+1)^2(s^2-s+1)(s^2+s+1)(2s^3-s^2+s-1)(2s^3+s^2+s+1). \tag{5}$$

The S^3 Coulomb graph at $n_{\max} = 3$ has three connected components indexed by $\ell = 0, 1, 2$. The $\ell = 0$ component is a 3-node path ($\beta_1 = 0$, $\zeta_{\ell=0} = 1$); the $\ell = 2$ component is a 5-node tree-like structure ($\beta_1 = 0$, $\zeta_{\ell=2} = 1$). *All non-trivial Ihara content lives in the $\ell = 1$ component*, a connected graph on 6 vertices (the $m \in \{-1, 0, 1\}$ states of the $n = 2, 3$ shells) with 7 edges — consistent with the degree-14 inverse zeta ($\deg \zeta_G^{-1} = 2E = 14$) and first Betti number $\beta_1 = E - V + 1 = 2$ (the two independent cycles visible in the factorisation). The cyclotomic factors $s^2 \pm s + 1$ in (5) have zeros at primitive sixth roots of unity (magnitude 1); the irreducible cubics $2s^3 \pm s^2 + s \pm 1$ have zeros at $|s| \approx 0.739$ (real) and $|s| \approx 0.823$ (complex), corresponding to longer closed walks through the component.

The per-ℓ decomposition is a direct consequence of the Fock adjacency preserving ℓ. It is an intrinsic structural property of the S^3 Coulomb graph and one of the reasons the S^3 Ihara zeta is analytically tractable.

5.2 S^5 Bargmann–Segal at $N_{\max} = 2$

$$\zeta_G(s)^{-1} \;=\; -(s-1)^6(s+1)^6(2s^2+1)^2(3s^4+3s^2+1)(24s^6+21s^4+s^2-1). \tag{6}$$

The graph is connected, $\beta_1 = 6$. The $(s \pm 1)^6$ prefactor captures the β_1 trivial zeros. The remaining factors are all even in s and have integer coefficients.

5.3 S^5 Bargmann–Segal at $N_{\max} = 3$: the $12 + 22$ dichotomy

$$\zeta_G(s)^{-1} \;=\; (s-1)^{23}(s+1)^{23} \cdot P_{12}(s) \cdot P_{22}(s), \tag{7}$$

where

$$\begin{aligned} P_{12}(s) &= 432s^{12} + 666s^{10} + 374s^8 + 135s^6 + 47s^4 + 11s^2 + 1, && (8)\\ P_{22}(s) &= 829440s^{22} + 2453184s^{20} + 3308104s^{18} + 2696682s^{16} \\ &\quad + 1470640s^{14} + 557227s^{12} + 146654s^{10} + 25709s^8 \\ &\quad + 2630s^6 + 79s^4 - 12s^2 - 1. && (9) \end{aligned}$$

Both P_{12} and P_{22} are polynomials in s^2 with integer coefficients; they together account for all $12 + 22 = 34$ non-trivial zeros, and combined with the $2 \cdot 23 = 46$ trivial zeros at $s = \pm 1$ give the Ihara polynomial degree $34 + 46 = 80$; the Hashimoto matrix has size $2|E| = 84$, the remaining 4 being zeros at $s = \infty$ from the degree-1 leaves.

The $12 + 22$ split of the Ihara zeta is combinatorially nontrivial: there is no obvious reason, from the adjacency alone, that the characteristic polynomial of T should factor into two irreducible integer polynomials of these specific degrees.

Observation 2 ($12 + 22$ factorization $= \mathbb{Z}_2$ m_ℓ-reflection block decomposition)**.** *The two factors $P_{12}(s^2), P_{22}(s^2)$ in the S^5 Bargmann–Segal $N_{\max} = 3$ Ihara zeta are exactly the characteristic polynomials of the node-level Ihara–Bass matrix $M(s) = I - sA + s^2Q$ restricted to the two blocks of the $\mathbb{Z}_2$ reflection $P :\; m_\ell \mapsto -m_\ell$. Explicitly,* $\det(M_{sym}) = (s-1)(s+1) \cdot P_{22}(s^2)$ *on the* 13-*dimensional symmetric block (containing the $m_\ell = 0$ fixed points), and* $\det(M_{antisym}) = P_{12}(s^2)$ *on the 7-dimensional antisymmetric block. The $\mathbb{Z}_2$ m_ℓ-reflection is the only sub-action of the Paper 25 [6] Hopf $U(1)$ that commutes with a real integer adjacency; the continuous $U(1)$ phase* *rotation reduces to this $\mathbb{Z}_2$ when the data is real. The analogous decomposition holds for the Dirac-S^3 Rule B $n_{\max} = 3$ graph, with the $22 + 24$ dichotomy corresponding to the equal-dimension $\mathbb{Z}_2$ blocks of the $m_j \to -m_j$ half-integer reflection (both blocks 14-dimensional because m_j is half-integer, so there are no reflection-fixed points).*

Observation 2 is validated by direct restriction of $M(s)$ to each reflection block in exact sympy rational arithmetic (`tests/test_hopf_u1_block.py`, Track RH-E): each block determinant equals the stated factor polynomial, and the product $\det(M_{\mathrm{sym}}) \cdot \det(M_{\mathrm{antisym}})$ equals the full $\det(M)$ symbolically.

Observation 3 (Magnetic reflection is Kramers; chirality reflection is Dirac-index–obstructed)**.** *The $m_j \to -m_j$ reflection of Observation 2 is a* fixed-point-free graph automorphism *of the Dirac-S^3 graph: its adjacency depends only on $|\Delta m_j|$, and half-integer m_j never vanishes, so every node is paired with a distinct partner — the combinatorial signature of Kramers degeneracy. Consequently the flat $\mathbb{Z}_2$ connections (spin structures, $H^1(G; \mathbb{F}_2)$) left invariant by this reflection form a large subspace of the cycle space (dimension 40 of 79 at $n_{\max} = 3$).*

The companion chirality *reflection $\chi \to -\chi$ — equivalently $\kappa \mapsto$ its opposite-sign partner at fixed ℓ — is by contrast not a graph automorphism at all, indeed not even a node bijection. At each orbital block (n, ℓ) the positive-chirality multiplet $\left(\kappa = -(\ell+1),\ j = \ell + \frac{1}{2}\right)$ carries $2\ell + 2$ states while the negative-chirality multiplet $\left(\kappa = +\ell,\ j = \ell - \frac{1}{2}\right)$ carries 2ℓ — a mismatch of $+2$ in* every *block, never cancelling. The total chirality imbalance is*

$$\#\{\chi = +1\} - \#\{\chi = -1\} \ = \ \sum_{(n,\ell)} 2 \ = \ n_{\max}(n_{\max} + 1),$$

the discrete spectral asymmetry (Dirac index) of the Camporesi–Higuchi spinor bundle. No permutation can swap sectors of unequal size, so the genuine-spin $\mathbb{Z}_2$ does not exist on the substrate, while the magnetic m_j reflection thrives.

Observation 3 was verified in exact $\mathbb{F}_2$ arithmetic (`tests/test_spin_structure_obstruction.py`, `debug/spin_structure_moduli_memo.md`): the m_j permutation is a fixed-point-free automorphism at $n_{\max} = 2, 3$, and the $(2\ell+2)$-vs-2ℓ block mismatch holds at $n_{\max} = 2, 3, 4$ with total $n_{\max}(n_{\max} + 1)$. It supplies the substrate-level reason behind the negative relativistic-$\mathbb{Z}_2$ tapering results (`debug/sprint_mj_pari` a chirality sign-count cannot stabilize a Hamiltonian when the underlying reflection is not a graph symmetry to begin with.

5.4 Algebraicity of zeros

Every factor in every closed form above has integer coefficients. The non-trivial Ihara zeros are therefore algebraic numbers, with Galois conjugates also appearing in the zero set. This is a direct consequence of the adjacency matrix being 0/1-valued and the Ihara–Bass formula (2) being a determinant of an integer-coefficient matrix polynomial.

Corollary 1 (π-freeness of Ihara zeros)**.** *No GeoVac Hopf Ihara zero has a closed form involving π, $\zeta(2)$, $\zeta(3)$, or any other transcendental from the Paper 18 taxonomy. Every zero is an algebraic number lying in a finite extension of $\mathbb{Q}$.*

Corollary 1 extends Paper 24's π-free certificate from adjacency data to Ihara-zeta data. The same bit-exact π-freeness is preserved under Ihara-zeta.

Corollary 2 (Integer-algebraicity of the reciprocal Ihara zeros)**.** *Every non-trivial* reciprocal *Ihara zero of a GeoVac Hopf graph — i.e. every nonzero eigenvalue of the Hashimoto edge operator T — is an algebraic integer (a root of a* monic *$\mathbb{Z}$-coefficient polynomial). The Ihara zeros themselves are algebraic over $\mathbb{Q}$ (Corollary 1) but are* not *in general algebraic integers.*

This sharpens Corollary 1 from "algebraic over $\mathbb{Q}$" to "algebraic integer" *for the reciprocal zeros*, and is direct from the Hashimoto formulation: T is a 0/1 edge-adjacency matrix, so its characteristic polynomial $\det(\lambda I - T)$ is *monic* with integer coefficients, and its eigenvalues (the reciprocal Ihara zeros) are therefore algebraic integers. The strengthening does *not* pass to the Ihara zeros themselves: they are the roots of the Ihara–Bass determinant $\det(I - sA + s^2Q)$, which has integer coefficients but leading coefficient $\det Q = \prod_v(\deg v - 1) \neq \pm 1$ in general — so the zeros are algebraic over $\mathbb{Q}$, not algebraic integers. (Concretely, the closed-form factor $2s^2 + 1$ of Section 5 has Ihara zeros $\pm i/\sqrt{2}$, algebraic but not algebraic integers; their reciprocals $\mp i\sqrt{2}$, roots of the monic $s^2 + 2$, are the T-eigenvalues and *are* algebraic integers.)

Observation 4 (Galois structure of Ihara zeros)**.** *Sprint 3 Track RH-F performed exact Galois analysis of the integer polynomial factors (5), (6), and (7), with these three classes of findings:*

1. *Cyclotomic factors $\Phi_3(s), \Phi_4(s), \Phi_6(s)$ appearing in the closed forms have Galois group $\mathbb{Z}/2$ and fingerprint specific graph cycles — Φ_3, Φ_6 track the 3-cycles and 6-cycles of the S^3 Coulomb $\ell = 1$ component (the $C_3 \times P_2$ ladder).*
2. *Ramanujan-boundary factors $(2s^2 + 1)^2$ (scalar S^5 $N_{\max} = 2$) and $(9s^2 + 1)^4$ (Dirac-S^3 Rule B $n_{\max} = 3$), whose zeros sit exactly on the critical Ramanujan circle $|s| = 1/\sqrt{q_{\max}}$, all live in $\mathbb{Q}(i)$. This is structurally consistent with the $\mathbb{Z}_2$ m-reflection sub-action of the Paper 25 Hopf $U(1)$ (Observation 2).*
3. *Bulk (non-boundary) factors have non-abelian Galois groups: S_3 for the scalar S^3 cubics; S_4 for the Dirac-A $n_{\max} = 3$ quartic; and S_6 (non-solvable) for the factor $P_{12}(s^2)$ in the scalar S^5 $N_{\max} = 3$ zeta. By Abel–Ruffini, the 12 zeros of P_{12} have no radical expression over $\mathbb{Q}$. By Kronecker–Weber, these factors do not lie inside any cyclotomic field $\mathbb{Q}(\zeta_M)$.*

In particular, the Hopf $U(1)$ structure controls the $\mathbb{Z}_2$ block split *(Observation 2) but does not abelianize the Galois group inside each block.*

Observation 4 adds a new taxonomic tier to Paper 18's exchange-constant hierarchy: *algebraic-but-non-radical*, sitting between radical-algebraic and transcendental. The S^5 $N_{\max} = 3$ P_{12} zeros are the first explicit GeoVac objects in this tier. Full per-factor Galois table in `debug/galois_ihara_memo.md`.

6 Functional Equation: the Irregular-Graph Reflection

For a $(q+1)$-regular graph, the Ihara zeta satisfies the Stark–Terras functional equation [17, 14]

$$\Xi_G(s) := (1 - s^2)^{r-1}\big(1 - (qs)^2\big)^{r-1}\zeta_G(s)^{-1}, \quad \Xi_G(s) = \Xi_G(1/(qs)). \tag{10}$$

In particular the Ihara zeros lie on the critical circle $|s| = 1/\sqrt{q}$ when the graph is Ramanujan. Both sanity witnesses $(K_4, K_{3,3})$ satisfy (10) with zeros on $|s| \in \{\frac{1}{\sqrt{2}}, 1\}$.

Neither GeoVac Hopf graph is regular: q varies from 0 (at leaf nodes) to $q_{\max}$ (at dense interior nodes). The Stark–Terras single-circle statement therefore does not hold; instead, the non-trivial Ihara zeros populate an *annulus* $1/\sqrt{q_{\max}} \leq |s| \leq 1$, and their distribution within the annulus is constrained only by the Hashimoto spectrum.

What *does* hold uniformly across our graphs:

Proposition 1 (Reflection symmetry). *For each GeoVac Hopf graph tested, $\zeta_G(s)^{-1}$ is invariant under $s \mapsto -s$ up to a possible sign factor of the trivial $(s \pm 1)^{\beta_1}$ prefactors. Equivalently, every non-trivial polynomial factor in the closed forms (5), (6), (7) is even in s.*

The proposition is a structural statement about undirected graphs: the Hashimoto matrix T satisfies $T \sim -T$ under the involution $e \mapsto e^{-1}$ on directed edges, so its characteristic polynomial is even in its argument. Proposition 1 is therefore a universal property of the Ihara zeta on any undirected graph, regular or not.

Proposition 2 (Algebraic closure under Galois conjugation). *The non-trivial Ihara zeros of every GeoVac Hopf graph are algebraic numbers whose Galois conjugates also appear in the zero set. The zero set is therefore a finite union of Galois orbits in $\bar{\mathbb{Q}}$.*

This is immediate from the integer-coefficient factorizations of Section 5.

The weakening from the full Stark–Terras symmetry to Propositions 1 and 2 is a feature of irregularity, not a pathology of our graphs. The correct Ξ-completion for the irregular case is expected to be a product over the degree strata of the graph, rather than a single regular-graph correction — the natural generalisation of the Stark–Terras regular-graph functional equation along the lines of the irregular Ihara–Bass machinery of Kotani–Sunada [13]. Making that completion explicit for our two graphs is a useful computational exercise but is not required to state the Ramanujan property, which is already secured by Observation 1.

7 Scope Boundary: No Natural Hyperbolic Quotient

A natural companion question to the Ihara-zeta result is whether the Wick-rotated continuation of the Fock projection, which takes the compact S^3 to the non-compact hyperbolic 3-space H^3 at positive energy [9], delivers a Selberg zeta with a *proven* RH analog (on congruence Bianchi quotients [19]). We investigated this in parallel and report a clean negative.

7.1 The Wick rotation is clean but continuous

Following Bander–Itzykson [9], the Fock projection with $p_0 = \sqrt{-2mE}$ continues to $p_0 = i\kappa_{\text{phys}}$ at $E > 0$, rotating the target manifold from the round $S^3 \subset \mathbb{R}^4$ to the upper unit pseudosphere in $\mathbb{R}^{3,1}$, i.e., hyperbolic 3-space H^3 in its Minkowski embedding. The symmetry group rotates from compact SO(4) to non-compact SO(3, 1), with the SO(4) principal-series representations continuing to the principal/complementary series of SO(3, 1). The Coulomb scattering amplitudes are matrix elements in this continuous principal series. This is clean and unambiguous.

7.2 But no discrete quotient $\Gamma \subset \mathrm{SO}(3,1)$ is singled out

To recover a graph-like discrete object on H^3, one needs a discrete subgroup Γ with quotient H^3/Γ. Of the four candidate Γ's that GeoVac structure suggests, none works:

1. **Weyl group of** SO(4)**.** The Weyl group $W(D_2) = \mathbb{Z}/2 \times \mathbb{Z}/2$ has order 4. Finite, not a lattice in SO(3, 1). Acting on H^3 it has fixed points everywhere; it does not produce a discrete hyperbolic quotient.

2. **Hopf S^1 fiber.** The compact Hopf S^1 is structurally irrelevant for H^3: hyperbolic 3-space has no Hopf-type principal bundle with a compact S^1 fiber. This is the same Coulomb/HO asymmetry Paper 24 [5] identified, one layer further.

3. **Bianchi groups** $\mathrm{PSL}(2, \mathcal{O}_K)$**,** $K = \mathbb{Q}(\sqrt{-d})$**.** The class-number-one cases $d \in \{1, 2, 3, 7, 11\}$ act discontinuously on H^3. None of GeoVac's numerical invariants – the Rydberg constant R_∞, the Fock momentum $p_0 = Z/n$, $\kappa = -\frac{1}{16}$, $B = 42$, $F = \pi^2/6$, $\Delta = \frac{1}{40}$, $\alpha^{-1} \approx 137$ – matches any specific d. The Bianchi groups are objects of number theory, not of hydrogen.

4. **Picard group** $\mathrm{PSL}(2, \mathbb{Z}[i])$**.** As the simplest Bianchi case, Picard is the default. But "simplest" is not a physics selection principle, and the Coulomb node labels $(n, \ell, m) \in \mathbb{Z}^3$ are not in $\mathbb{Z}[i]$. No natural selection.

The literature survey [20] that accompanied this investigation confirmed that *no published work* links the SO(4) Coulomb symmetry to Bianchi arithmetic quotients.

Observation 5 (No natural hydrogen-Selberg bridge)**.** *The Fock Wick rotation $S^3 \to H^3$ is well-defined and continuous, but GeoVac provides no selection principle for a discrete subgroup $\Gamma \subset$* $\mathrm{SO}(3, 1)$*. The Selberg-zeta angle on hydrogen-derived hyperbolic quotients is therefore structurally closed.*

This observation refines the Paper 24 [5] universal / Coulomb-specific partition one level further:

- S^3 (Coulomb, bound): compact, Hopf, calibration π, discrete spectrum, graph-zeta available.
- S^5 (HO, Bargmann–Segal): compact complex, Hopf $U(1)$, bit-exactly π-free, discrete spectrum, graph-zeta available.
- H^3 (Coulomb, continuum): non-compact, no natural Hopf bundle, no natural discrete subgroup, continuous spectrum, Selberg-zeta *requires external arithmetic input.*

The graph-zeta direction is therefore the single RH-adjacent direction that is intrinsic to GeoVac. The Selberg direction, for hydrogen, is not.

8 Open Questions

Extension to $N_{\max} = 5$. The Paper 25 headline case for the S^5 Bargmann–Segal graph reports $V = 56$, $E = 165$, $\beta_1 = 110$. The Hashimoto operator at that size is $2|E| = 330$-dimensional – a fully tractable numerical eigensolve. Extending the Ramanujan verdict and the closed-form factorization to $N_{\max} = 5$ would give three data points on the deviation's growth (already at -0.357 for $N_{\max} = 3$) and a headline match to Paper 25's Hodge data.

Alon–Boppana asymptotics — resolved (Sprint 2, RH-D). Extending the Ramanujan verdict to $n_{\max}, N_{\max} \in \{4, 5\}$ in a post-draft sweep establishes that the signed deviation grows *linearly in V* for three of four families (scalar S^3 Coulomb: $R^2 = 0.994$; scalar S^5 Bargmann–Segal: $R^2 = 0.766$; Dirac-S^3 Rule B: $R^2 = 0.828$). The Kotani–Sunada bound is crossed between $V \approx 30$ and $V \approx 60$ for each of these families: scalar S^3 at $n_{\max} = 5$ ($V = 55$, deviation $+0.20$), scalar S^5 at $N_{\max} = 5$ ($V = 56$, deviation $+0.65$), Dirac-S^3 Rule B at $n_{\max} = 4$ ($V = 60$, deviation $+1.53$). Only the Dirac-S^3 Rule A family appears to remain sub-Ramanujan, with a $1/V$-class envelope ($|\text{deviation}| \sim V^{-3.05}$, $R^2 = 0.994$). The conjectured structural mechanism is *mixed-density topology*: $q_{\max}$ is set by the single densest vertex while the Perron eigenvalue of T is driven by graph volume; as the graph grows, near-regular dense sub-blocks appear whose Perron exceeds $\sqrt{q_{\max}}$. The graph-RH for GeoVac is a finite-size statement, secured for $n_{\max}, N_{\max} \leq 3$ and failing asymptotically for the three dense families. The finite-size sign flip (Ramanujan at small size, crossing at $V \approx 30$–60) is recomputed from the live graph spectra in `tests/test_paper29_bound_crossing.py`; full data at `debug/data/alon_boppana_sweep.json`.

Hopf-quotient block decomposition — validated (Sprint 2, RH-E). Observation 2 has been verified by direct restriction of the node-level Ihara–Bass matrix $M(s)$ to each $\mathbb{Z}_2$ m-reflection block in exact sympy rational arithmetic. The block decomposition reproduces P_{12} and P_{22} (scalar S^5 $N_{\max} = 3$) and P_{22} and P_{24} (Dirac-S^3 Rule B $n_{\max} = 3$) symbolically. The $\mathbb{Z}_2$ reflection is the only sub-action of the Paper 25 Hopf $U(1)$ that commutes with a real integer adjacency, and the mechanism is identical in scalar and spinor cases. The Matsuura–Ohta Kazakov–Migdal / Ihara-zeta program on generic finite graphs [21, 22] provides a parallel direction in which gauge-theoretic duality structures on the graph may shadow-match the Hopf-$U(1)$ block decomposition recorded here; the connection is noted as a forward pointer, not pursued in the present work.

Spin-ful Ihara zeta on the Dirac-S^3 graph. The infrastructure built for the Tier-3 relativistic sprint [3, 4] – the Dirac spectrum $|\lambda_n| = n + \frac{3}{2}$, the (κ, m_j)-labeled spinor harmonic basis, the Szmytkowski matrix elements – induces a distinct graph structure on the Dirac sector. Its Ihara zeta is a genuinely new object: the edges differ from the Schrödinger graph (the Dirac rule couples spin and orbital angular momentum), and the resulting Hashimoto spectrum is – conjecturally – richer. This is the natural spinor extension of the present paper and the recommended next Ihara-zeta sprint from both the Wick memo and the present work.

Continuum limit — redirected from Ihara to spectral zeta (Sprint 2, RH-G). Sprint 2 Track RH-G investigated whether any combinatorial or refinement limit of the GeoVac Ihara zeta connects to a classical L-function. The headline obstruction is structural: the GeoVac Hopf graphs are *bipartite* (all closed non-backtracking walks have even length), so $\mathrm{tr}(T^L) = 0$ for odd L and the Ihara walk-length spectrum is supported on the even integers. Paper 28's χ_{-4} Dirichlet character has support only on odd integers, so the χ_{-4}-twisted Ihara zeta $\zeta_G(s; \chi_{-4}) = \prod_{[C]}(1 - \chi_{-4}(\ell(C)) s^{\ell(C)})^{-1}$ is identically the trivial Euler product. The χ_{-4} structure that does appear in Paper 28 lives on the *spectral* side — the Dirac Dirichlet series $D(s) = \sum_n g_n \cdot |\lambda_n|^{-s}$ at quarter-integer Hurwitz shifts, splitting into $D_{\mathrm{even}}(s)$ and $D_{\mathrm{odd}}(s)$ whose differences expose Catalan's $G = \beta(2)$ and $\beta(4)$. Supporting statistics: the Hashimoto eigenvalue-spacing coefficient of variation is ≈ 1 (Poisson, Berry–Tabor), not the GUE value ≈ 0.42, categorically excluding a direct Hilbert–Pólya identification with ζ zeros. The bridge to classical $\zeta(s)$, if any, must therefore go through the *spectral* zeta (Paper 28's machinery), not through the Ihara zeta. Full obstruction analysis at `debug/riemann_limit_memo.md`.

Spectral-side GUE signature (Sprint 3, RH-M). Sprint 3 Track RH-M computed the complex zeros of $D(s)$, $D_{\mathrm{even}}(s)$, and $D_{\mathrm{odd}}(s)$ numerically in the strip $\mathrm{Im}(s) \in [0, 70]$ at 35-digit precision, and tested the unfolded imaginary-part spacings against GUE and Poisson reference statistics. All three spectral Dirichlet series show *GUE-like* coefficient of variation $\mathrm{CV} \in \{0.348, 0.343, 0.396\}$, compared to synthetic GUE $\mathrm{CV}_{\mathrm{GUE}} = 0.46$ and synthetic Poisson $\mathrm{CV}_{\mathrm{Poisson}} = 1.05$. Kolmogorov–Smirnov testing against Poisson rejects ($p = 0.004, 0.030$ for $D_{\mathrm{even}}, D_{\mathrm{odd}}$); against GUE does not reject ($p = 0.61, 0.65, 0.94$). This is the *first* GUE signature in the GeoVac framework, and it is on exactly the side that the Sprint 2 RH-G redirection pointed to — the spectral Dirichlet side, not the Ihara side (whose Hashimoto spacing is Poisson, CV ~ 1, per Sprint 2 RH-G). Structurally: the two sides of the Weil dictionary give opposite verdicts on GeoVac. The spectral side is Hilbert–Pólya compatible; the Ihara side is not. Important honest caveat: the zeros are *not* on a single critical line (they are scattered in $\mathrm{Re}(s) \in [2.02, 3.42]$), so the GUE signature is spacing-only, not an RH-style line constraint. Classical ζ has RMT spacing *and* zeros-on-$\Re s = 1/2$ simultaneously; GeoVac has only the first. Full sample-size caveats, KS p-values, and pair-correlation data in `debug/spectral_zero_stats_memo.md`. Cross-sprint coherence: combined with Sprint 3 RH-J's closed-form identity $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ (Paper 28 [7], recent subsection), the χ_{-4} Dirichlet-L content sits inside a GUE-compatible spectral object.

α^2-weighted Ihara zeta on the Dirac-S^3 graph (Sprint 3, RH-K). The unweighted Ihara zeta depends only on 0/1 connectivity. Sprint 3 Track RH-K introduced a non-trivial edge weighting $w(a,b) = 1 + \alpha^2 \cdot |f_{\text{SO}}(a) + f_{\text{SO}}(b)|/2$ with $f_{\text{SO}}(n,\kappa) = -(\kappa+1)/[4n^3\ell(\ell+\frac{1}{2})(\ell+1)]$, the dimensionless Breit–Pauli spin-orbit diagonal. The weighted Ihara zeta $\zeta_G^w(s)$ has well-defined Ihara–Bass and Hashimoto analogs (Mizuno–Sato 2004, Setyadi–Storm 2022). Four principal findings: (i) the weighted zeta lives in the ring $\mathbb{Q}(\alpha^2)[s]$, an explicit realization of Paper 18's spinor-intrinsic tier without the γ factor; (ii) α^2 enters the zero set as a *graph-invariant*, not just an edge-local perturbation — Rule A $n_{\max} = 3$ gives ζ^{-1} as a polynomial in α up to degree 44, with the α^2 coefficient carrying a clean rational prefactor $-35/648$; (iii) weighting does not shift Hashimoto spacing toward GUE — in fact CV grows away from GUE under weighting (Rule B $n_{\max} = 3$: $2.6 \to 5.5$), closing the Ihara-side Hilbert-Pólya route further; (iv) the boundary-Ramanujan case (Rule B $n_{\max} = 2$, unweighted deviation exactly zero) crosses the bound under α^2-weighting for any $\alpha > 0$, but this is a generic perturbation-of-boundary effect, not an α^2-specific signature. The weighted Ihara zeta is therefore a new algebraic object (first GeoVac graph observable with genuine α^2 content), with structural implications for Paper 18's taxonomy and no immediate RH-adjacent use. Full analysis in `debug/ihara_zeta_weighted_memo.md`; module `geovac/ihara_zeta_weighted.py`.

Hilbert–Pólya operator reconstruction (Sprint 4, RH-N). Given the GUE-like zeros of $D(s)$, D_{even}, D_{odd} (RH-M), a Hilbert–Pólya operator $H^\star$ — a Hermitian matrix whose eigenvalues are $\{\text{Im}(\rho_n)\}$ — can be explicitly reconstructed from the zero list. Four independent constructions (diagonal, tridiagonal Jacobi via Lanczos, Toeplitz via DCT, companion-matrix unitary-conjugate) all reproduce the spectrum; cross-construction Frobenius distances 0.40–0.81 quantify that eigenvalues alone do not determine the operator. The structural headline is a *Weyl law* observed across all three zero sets: $\gamma_n \approx 18.5\sqrt{n} + \text{const}$, with RMS residual 0.8–0.9 and outperforming both linear (2×–4×) and $n \log n$ (3×–5×) fits. This $\sqrt{n}$ growth is the Weyl density of a *1D harmonic-oscillator-like* system, categorically different from the linear $n + 3/2$ growth of the Dirac operator on S^3 and from the log-density of classical Riemann zeros. $H^\star$ is therefore *not* a small deformation of any native GeoVac Hamiltonian. Module `geovac/hp_operator.py` (13 tests); full analysis in `debug/hp_operator_memo.md`.

Functional equation hunt on the spectral Dirichlet series (Sprint 4, RH-O). A systematic grid search was performed over 13,080 candidate completed-ξ templates of the form $\xi_G(s) = P(s) \cdot \rho^s \cdot \prod_i \Gamma((s + a_i)/b_i) \cdot G(s)$, covering 8 prefactors, 9 scale parameters ρ, 30 single-Gamma factors, and 55 Gamma pairs, tested at 10 complex points against 15 candidate reflection axes, in mpmath 50-digit precision. The sanity anchors (classical ξ_ζ, ξ_β, and the RH-J identity itself) all close to residual $\sim 10^{-51}$. The global best residual across all GeoVac templates and all axes is 0.594 — 48 orders of magnitude worse than the sanity threshold. The search therefore produces a clean negative. The structural obstruction: each of $D(s)$, $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$ is a linear combination of *two* Hurwitz pieces at *different exponents* s and $s - 2$ with mismatched weights, and no single Gamma product can simultaneously complete both; for D_{diff}, the two β pieces require different reflection axes ($c = 1$ for $\beta(s)$, $c = 5$ for $\beta(s-2)$). The missing critical-line-forcing ingredient of classical RH is therefore *structurally* absent from GeoVac's spectral zeta, not merely undiscovered. Combined with RH-M's non-critical-line zero distribution and RH-N's $\sqrt{n}$ Weyl law, Sprint 4 closes the direct spectral-to-classical-RH bridge. Full analysis in `debug/functional_equation_hunt_memo.md`.

Depth-2 χ_{-4} analog for $S_{\min}$ (Sprint 4, RH-P). The depth-1 identity of Paper 28 [7] has no depth-2 analog for the CG-weighted irreducible constant $S_{\min}$. Splitting $S_{\min} = \sum_k T(k)^2$ by the parity of k gives $S_{\min}^{\text{even}} + S_{\min}^{\text{odd}} = S_{\min}$ (sanity); the difference $S_{\min}^{\text{diff}}$ is numerically irreducible against χ_{-4} / $\beta(s)$ / $\zeta(s)$ bases up to weight 8 (35 PSLQ attempts across 7 basis strategies at 100 dps, zero identifications). The structural obstruction: RH-J.1's telescoping relied on the *linear* Hurwitz identity $\zeta(s, 3/4) - \zeta(s, 5/4) = 4^s(1 - \beta(s))$ with uniform 2^{2s-1} prefactors. At depth 2, $T(k)^2$ is a *quadratic* object; the Bernoulli-asymptotic tail coefficients scramble the uniform prefactor structure, and $S_{\min}^{\text{even}}$, $S_{\min}^{\text{odd}}$ live in a weight-2 multiple-zeta-value sector not spanned by products of standard Dirichlet-L values. Full analysis in `debug/smin_chi_neg4_memo.md`. A side-finding concerning the published numerical value of $S_{\min}$ is under independent verification at the time of writing; the irreducibility claim is unaffected either way.

SU(2) Wilson lattice gauge on $S^3 = \mathrm{SU}(2)$ (Sprint 4, RH-Q). A self-consistent non-abelian Wilson lattice gauge theory with gauge group SU(2) was constructed on the Fock-projected S^3 Coulomb Hopf graph, extending Paper 25's $U(1)$ Wilson–Hodge structure. Link variables $U_e \in \mathrm{SU}(2)$ on directed edges; plaquettes are primitive closed non-backtracking walks (our Ihara cycles); action $S_W = \beta \sum_{\text{plaq}} (1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} U_{\text{plaq}})$; character expansion via SU(2) Haar. Three structural results: (i) Paper 25's $U(1)$ action is *exactly* the maximal-torus limit of this SU(2) theory (diagonal U_e reduce to $U(1)$ to machine precision), elevating the abelian construction to a Cartan projection of a genuinely non-abelian theory; (ii) weak-coupling linearization around the trivial vacuum yields exactly the discrete Hodge-1 Laplacian $L_1 = B^T B$ of Paper 25 as the kinetic term, retroactively identifying L_1 as the SU(2) kinetic sector; (iii) plaquette counts grow $(0, 0, 0) \to (2, 1, 1) \to (8, 7, 18)$ at length $(L = 4, 6, 8)$ for $n_{\max} = 2, 3, 4$; Monte-Carlo Wilson loops on $n_{\max} = 3$ at $\beta \in \{0.5, 1, 2, 5\}$ give $\langle W \rangle = 0.126, 0.158, 0.332, 0.681$, monotonic but without an area/perimeter-law crossover at this size (only one plaquette class exists). This is a gauge-invariant, Haar-normalizable non-abelian lattice gauge theory on a compact finite graph; it is *not* a continuum Yang–Mills mass-gap result (the Millennium Problem is specifically about $\mathbb{R}^4$ continuum QFT with a non-abelian simple gauge group; our construction is a discrete compact-graph version). Module `geovac/su2_wilson_gauge.py` (26 tests); full analysis in `debug/su2_wilson_gauge_memo.md`.

9 Conclusion

The GeoVac Hopf graphs satisfy the graph Riemann Hypothesis — the Kotani–Sunada Ramanujan bound — strictly, at every small size tested ($n_{\max}, N_{\max} \le 3$). Extending to $n_{\max}, N_{\max} \in \{4, 5\}$ in a Sprint-2 follow-up sweep shows that three of four graph families cross the bound at modest sizes ($V \approx 30$–60) with a signed deviation growing linearly in V, so the graph-RH is a finite-size statement for GeoVac, not an asymptotic one (Section 8, Alon–Boppana subsection). Their Ihara zetas factor with integer coefficients; their zeros are algebraic; no transcendentals enter the zero set, consistent with Paper 24's π-free certificate. The per-ℓ-shell decomposition of the S^3 Coulomb zeta and the $12 + 22$ / $22 + 24$ dichotomies of the scalar S^5 $N_{\max} = 3$ and Dirac-S^3 Rule B $n_{\max} = 3$ zetas are, by direct verification, the $\mathbb{Z}_2$ m-reflection blocks of the node-level Ihara–Bass matrix (Observation 2); these connect back to Paper 7's SO(4) selection rules and to the real-integer-adjacency-compatible sub-action of Paper 25's Hopf-$U(1)$ gauge structure. Neither graph is regular, so the single-critical-circle Stark–Terras functional equation does not apply; what does hold uniformly is the $s \mapsto -s$ reflection and Galois closure of the zero set.

The adjacent Selberg-zeta direction – a Wick rotation of the Fock projection to hyperbolic 3-space, followed by a quotient by a discrete arithmetic subgroup – is structurally closed for hydrogen: the framework supplies no selection principle for such a subgroup. The graph-zeta angle is therefore the single RH-adjacent direction intrinsic to GeoVac.

We frame this as an observation rather than a claim about the classical Riemann Hypothesis. The graph-RH for a specific family of finite graphs is a different object than RH for $\zeta(s)$. What these results establish is a concrete bridge at small graph sizes: *our* framework has an Euler product, a functional equation (in the irregular-graph form), a critical region, and zeros that respect the Ramanujan bound at $n_{\max}, N_{\max} \leq 3$, plus a validated $\mathbb{Z}_2$ block-decomposition matching the Paper 25 Hopf structure. The Sprint 2 search for a limiting bridge to classical $\zeta(s)$ (Track RH-G) produced a structural redirection: the bipartiteness of the GeoVac Hopf graphs blocks any Ihara-side bridge, but Paper 28's spectral Dirichlet series (equipped with χ_{-4}) remains the open direction on the spectral side. The Ihara zeta therefore terminates as a self-contained structural object with validated $\mathbb{Z}_2$ block decomposition, and the Prize-bound research program moves to the spectral-zeta side.

The computations reported here are recorded in `geovac/ihara_zeta.py` and `tests/test_ihara_zeta.py`, with full numerical data in `debug/data/ihara_zeta_geovac_hopf.json` and the supporting Wick-rotation memo at `debug/fock_continuation_memo.md`. All Ihara zeta closed forms above are exact in `sympy` rational arithmetic.

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